

Bartonella jaculi sp. nov., *Bartonella callosciuri* sp. nov., *Bartonella pachyuromydis* sp. nov. and *Bartonella acomydis* sp. nov., isolated from wild Rodentia

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Four novel strains of members of the genus *Bartonella*, OY2-1^T, BR11-1^T, FN15-2^T and KS2-1^T, were isolated from the blood of wild-captured greater Egyptian jerboa (*Jaculus orientalis*), plantain squirrel (*Callosciurus notatus*), fat-tailed gerbil (*Pachyuromys duprasi*) and golden spiny mouse (*Acomys russatus*). All the animals were imported to Japan as pets from Egypt, Thailand and the Netherlands. The phenotypic characterization (growth conditions, incubation periods, biochemical properties and cell morphologies), DNA G+C contents (37.4 mol% for strain OY2-1^T, 35.5 mol% for strain BR11-1^T, 35.7 mol% for strain FN15-2^T and 37.2 mol% for strain KS2-1^T), and sequence analyses of the 16S rRNA genes indicated that those strains belong to the genus *Bartonella*. Sequence comparisons of *gltA* and *rpoB* genes suggested that all of the strains should be classified as novel species of the genus *Bartonella*. In phylogenetic trees based on the concatenated sequences of five loci, including the 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and the ITS region, and on the concatenated deduced amino acid sequences of three housekeeping genes (*ftsZ*, *gltA* and *rpoB*), all strains formed distinct clades and had unique mammalian hosts that could be discriminated from other known species of the genus *Bartonella*. These data strongly support the hypothesis that strains OY2-1^T, BR11-1^T, FN15-2^T and KS2-1^T should be classified as representing novel species of the genus *Bartonella*. The names *Bartonella jaculi* sp. nov., *Bartonella callosciuri* sp. nov., *Bartonella pachyuromydis* sp. nov. and *Bartonella acomydis* sp. nov. are proposed for these novel species. Type strains of *Bartonella jaculi* sp. nov., *Bartonella callosciuri* sp. nov., *Bartonella pachyuromydis* sp. nov. and *Bartonella acomydis* sp. nov. are OY2-1^T (=JCM 17712^T=KCTC 23655^T), BR11-1^T (=JCM 17709^T=KCTC 23909^T), FN15-2^T (=JCM 17714^T=KCTC 23657^T) and KS2-1^T (=JCM 17706^T=KCTC 23907^T), respectively.

The GenBank/EMBL/DDBJ accession numbers for the partial sequences of the 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and ITS regions of the four type strains described in this paper are as follows: *Bartonella jaculi* OY2-1^T, AB602527, AB602539, AB444975, AB529934 and AB602557, respectively; *Bartonella callosciuri* BR11-1^T, AB602530, AB602542, AB602551, AB529931 and AB602560, respectively; *Bartonella pachyuromydis* FN15-2^T, AB602531, AB602543, AB444978, AB602555 and AB602561, respectively; *Bartonella acomydis* KS2-1^T, AB602533, AB602545, AB444979, AB529942 and AB602563, respectively. Accession numbers for partial sequences of the above five loci in the other four isolates described in this paper (OY5-1, BR1-1, FN18-1 and KS7-1) are summarized in Table 1.

A supplementary figure and supplementary table are available with the online version of this paper.

The genus *Bartonella* is classified in the class *Alpha-proteobacteria*, order *Rhizobiales* and family *Bartonellaceae*, and at the time of writing consists of 24 species and three subspecies. The bacteria in the genus *Bartonella* show fastidious, Gram-negative, aerobic and slow-growing properties. *Bartonella* strains have been documented to infect erythrocytes of various mammals, especially rodents in North America (Kosoy *et al.*, 1997), Asia (Ying *et al.*, 2002; Inoue *et al.*, 2008; Inoue *et al.*, 2010), Australia (Gundi *et al.*, 2009) and Europe (Birtles *et al.*, 1995; Tea *et al.*, 2004). We have already reported that rodents and squirrels imported to Japan as pets from Asia, North America, Europe and the Middle and Near East carried organisms of the genus *Bartonella* at a high prevalence (26.0%). In the

previous study, a total of 407 isolates were obtained from the 142 bacteraemic animals and the isolates were classified into 10 genogroups consisting of four zoonotic known species of the genus *Bartonella* and six novel species of the genus *Bartonella* according to the phylogenetic analysis based on the sequence of *gltA* gene (Inoue *et al.*, 2009). The purpose of the present study was to further characterize four of the six novel species of the genus *Bartonella* by biochemical, morphological and genetic approaches, including multilocus sequencing analysis of four housekeeping genes and the 16S–23S rRNA intergenic spacer region. Furthermore, we compared the specific hosts of the isolates with species of the genus *Bartonella* with validly published names because Bartonellae have evolved towards adaptation to specific mammalian hosts (Chomel *et al.*, 2009).

Four novel species of the genus *Bartonella* were isolated from the rodents and squirrels from Egypt, Thailand and the Netherlands, and two strains from each animal species were selected from the each clade in the phylogenetic tree of the *gltA* gene for further analysis. The strains and their animal sources are summarized in Table 1. Each strain was grown on heart infusion agar plates (Difco) containing 5% defibrinated rabbit blood (HIA) at 35 °C in a moist atmosphere under 5% CO₂ for 14 days. After Gram staining, each strain was observed by light microscopy (Olympus) at ×1000 magnification. The surface structures of the cells were analysed by transmission electron microscopy (model JEM1200EX; JEOL) at 100 kV with negative staining. Cytochrome oxidase test strips (Nissui) were used for evaluating the oxidase activity of the bacteria. Catalase activity was examined by mixing fresh colonies that had been cultured on rabbit blood–chocolate HIA plates for 14 days at 35 °C in a moist atmosphere containing 5% CO₂ with 3% H₂O₂ on a glass slide. A total of 23 biochemical characteristics were assessed by using a MicroScan Rapid Anaerobe Panel (Dade Behring) according to the manufacturer's instructions as previously described (Welch *et al.*, 1993).

Genomic DNA was extracted from each strain by using the Instagene Matrix (Bio-Rad) according to the manufacturer's instructions. Partial sequences of the 16S rRNA-encoding gene (16S rRNA), the cell division protein gene (*ftsZ*), the citrate synthase gene (*gltA*), the RNA polymerase beta-subunit-encoding gene (*rpoB*) and 16S–23S rRNA intergenic spacer region (ITS region) were amplified by PCR as described previously (Birtles & Raoult, 1996; Heller *et al.*, 1997; Renesto *et al.*, 2001; Houpiakian & Raoult, 2001; Zeaiter *et al.*, 2002). The sequence data were submitted to GenBank and the accession numbers were obtained (Table 1). In addition, the full-length sequence of another cell division protein gene (*ftsY*) was determined and the DNA G+C content (mol%) of each strain was estimated by the sequence as previously described (Fournier *et al.*, 2006; Fournier *et al.*, 2007; Mediannikov *et al.*, 2008). The primers for amplification of the full-length *ftsY* gene were originally designed in this study (forward primer, 5'-CHGGAACAATTGTTMAAGC-3'; reverse primer, 5'-CYAATGGHCCCATTTTCYAAG-3'; sequencing primer 1,

5'-ACTAGGDGCAGATGCGC-3'; sequencing primer 2, 5'-TYGGCAAAGTCAGARGC-3'). PCR amplification for *ftsY* gene was performed as follows: a 5 min denaturation at 95 °C was followed by 35 cycles of denaturation for 30 s at 94 °C, annealing for 30 s at 47 °C and extension for 60 s at 72 °C.

PCR products were purified by using a Spin Column PCR product purification kit (Bio Basic) then sequenced directly by using dye terminator chemistry and a Genetic Analyzer model 3130 (Applied Biosystems).

For the phylogenetic analysis, the sequence data for the 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and the ITS region of the strains were aligned and subsequently compared with those of type strains of known species of the genus *Bartonella* from GenBank by using the CLUSTAL W program in MEGA 4.0.2 software (Tamura *et al.*, 2007). The phylogenetic tree was constructed by the neighbour-joining method from the concatenated sequences of the 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and the ITS region with the Jukes–Cantor parameters distance model. The phylogenetic tree of the concatenated deduced amino acid sequences of the *ftsZ*, *gltA* and *rpoB* gene products was also constructed by using the neighbour-joining method and the evolutionary distances were computed using the Poisson correction method. Support for the nodes in the trees was assessed by bootstrapping with 1000 replicates. *Brucella melitensis* 16M^T was used as an outgroup.

All of the strains grown on HIA formed smooth, transparent to grey–whitish colonies of 1–2 mm in diameter. Colonies of most strains were round, but colonies of strains KS2-1^T and KS7-1 were irregular. Gram-negative, coccobacilli to short rod-shaped bacteria were observed for all of the strains by light microscopy after 14 days culture. Electron microscopic examination showed that strains OY2-1^T, BR11-1^T, FN15-2^T and KS2-1^T were similar in cell size: the length ranging from 1.15 to 1.45 µm and width ranging from 0.52 to 0.59 µm, respectively (Fig. 1). The strains had no flagella, but pili-like structures around the cell body were observed in the four strains (Fig. 2).

All of the strains were negative for oxidase, catalase and urease activities. Nitrate reduction, indole production and indoxyl phosphate degradation were not observed in any of the strains. None of the strains hydrolysed trehalose and most of *p*-nitrophenyl substrates: *N*-acetyl-β-D-glucosaminide, α-D-galactopyranoside, β-D-galactopyranoside, α-D-glucopyranoside, β-D-glucopyranoside, α-L-fucopyranoside and α-D-mannopyranoside. Strains FN15-2^T, FN18-1, KS2-1^T and KS7-1 hydrolysed bis-*p*-nitrophenyl phosphate, but strains OY2-1^T, OY5-1, BR1-1 and BR11-1^T did not. Amino acid arylamidase activities towards leucine, methionine, lysine (alkaline as well as acidic), glycine, glycyglycine, arginine and tryptophan were observed in all the strains, but there was no activity for pyrrolidonyl. All of the strains except BR1-1 and BR11-1^T showed the ability to degrade L-proline-β-naphthylamide. The results of these biochemical tests are similar to those for other members of

Table 1. Host animals, GenBank accession numbers and sequence similarities (%) of five loci in four novel species of the genus *Bartonella*

Abbreviations: B. a, *B. alsatica*; B. b, *B. birtlesii*; B. c, *Bartonella clarridgeiae*; B. d, *B. doshiae*; B. e, *B. elizabethae*; B. g, *B. grahamii*; B. h, *B. henselae*; B. k, *Bartonella koehlerae*; B. que, *Bartonella queenslandensis*; B. qt, *Bartonella quintana*; B. r, *B. rattaaustraliani*; B. s, *B. silvatica*; B. tay, *Bartonella taylorii*; B. tri, *B. tribocorum*; B. v. a, *Bartonella vinsonii* subsp. *arupensis*; B. v. v, *Bartonella vinsonii* subsp. *vinsonii*.

Species	Host (scientific name)	Strain	GenBank accession number [most closely related species of the genus (sequence similarity, %)]				
			16S rRNA (1363 bp)	<i>ftsZ</i> (788 bp)	<i>gltA</i> (312 bp)	<i>rpoB</i> (825 bp)	ITS region (1251 bp)
<i>Bartonella jaculi</i>	Greater Egyptian jerboa (<i>Jaculus orientalis</i>)	OY2-1 ^T	AB602527 [B. k (99.2)]	AB602539 [B. v. v (92.5)]	AB444975 [B. v. a (93.6)]	AB529934 [B. qt (90.9)]	AB602557 [B. tay (75.3)]
		OY5-1	AB602528 [B. k (99.0)]	AB602540 [B. v. v (92.5)]	AB444976 [B. v. a (93.3)]	AB602554 [B. qt (90.9)]	AB602558 [B. tay (75.3)]
<i>Bartonella callosciuri</i>	Plantain squirrel (<i>Callosciurus notatus</i>)	BR1-1	AB602529 [B. que, B. s (99.3)]	AB602541 [B. a (93.3)]	AB444977 [B. v. v (94.6)]	AB529929 [B. a (91.5)]	AB602559 [B. v. v (71.8)]
		BR11-1 ^T	AB602530 [B. que, B. s (99.3)]	AB602542 [B. a (93.3)]	AB602551 [B. v. v (94.6)]	AB529931 [B. a (91.5)]	AB602560 [B. v. v (71.8)]
<i>Bartonella pachyuromydis</i>	Fat-tail gerbil (<i>Pachyuromys duprasi</i>)	FN15-2 ^T	AB602531 [B. tri (99.4)]	AB602543 [B. a (93.1)]	AB444978 [B. b, B. v. v (91.3)]	AB602555 [B. v. v (91.7)]	AB602561 [B. d (70.9)]
		FN18-1	AB602532 [B. que (99.3)]	AB602544 [B. a (92.3)]	AB602552 [B. b, B. v. v (91.3)]	AB529949 [B. v. v (91.7)]	AB602562 [B. d (71.6)]
<i>Bartonella acomydis</i>	Golden spiny mouse (<i>Acomys russatus</i>)	KS2-1 ^T	AB602533 [B. r (99.3)]	AB602545 [B. a (92.8)]	AB444979 [B. b (92.6)]	AB529942 [B. b (93.4)]	AB602563 [B. e (62.7)]
		KS7-1	AB602534 [B. r (99.3)]	AB602546 [B. a (92.9)]	AB602553 [B. b (92.6)]	AB602556 [B. b (93.4)]	AB602564 [B. e (62.9)]

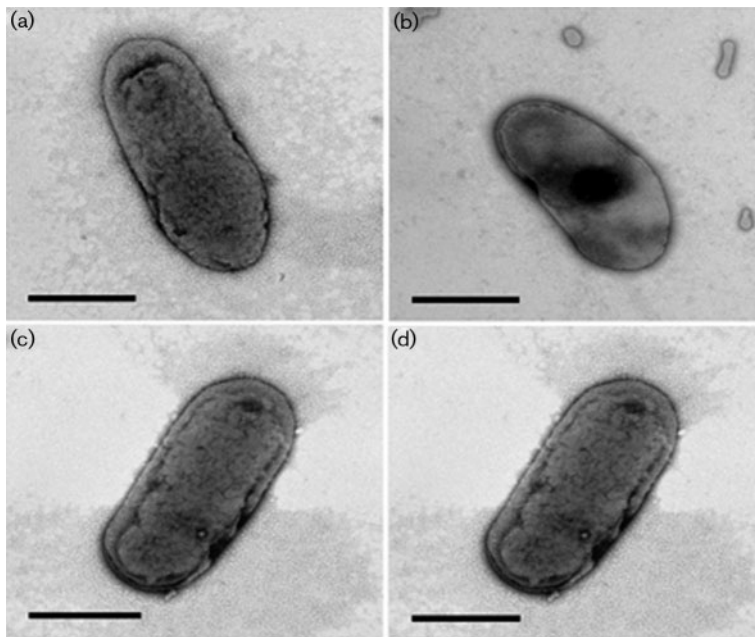


Fig. 1. Transmission electron micrographs of negatively stained cells of (a) *Bartonella jaculi* sp. nov. strain OY2-1^T, (b) *Bartonella callosciuri* sp. nov. strain BR1-1^T, (c) *Bartonella pachyuromydis* sp. nov. strain FN15-2^T and (d) *Bartonella acomydis* sp. nov. strain KS2-1^T. Bars, 500 nm.

the genus *Bartonella* (Table S1, available in IJSEM online); however, the profiles cannot be used for the differentiation of species of the genus *Bartonella* because of the relatively inert nature of Bartonellae (Dehio *et al.*, 2001; Bermond *et al.*, 2002). All of the morphological and biochemical characteristics of the four novel *Bartonella* species are summarized in Table S1.

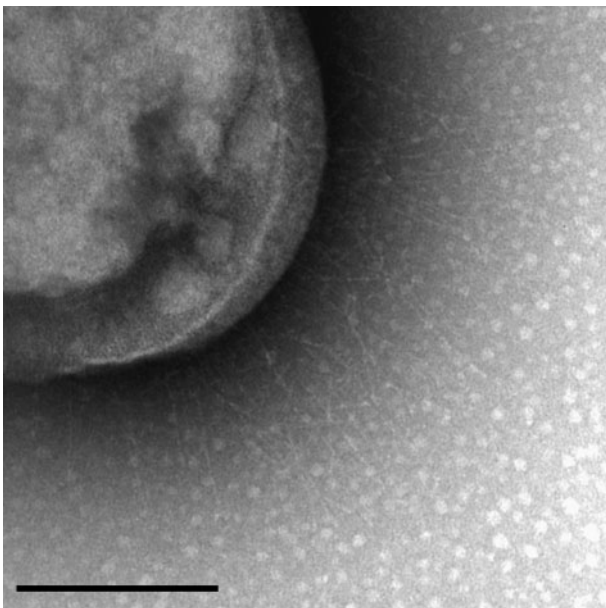


Fig. 2. Transmission electron micrograph showing the pili-like structures around the cell body of strain FN15-2^T. The structures were confirmed in all strains tested. Bar, 200 nm.

The G+C contents of the *ftsY* sequences ranged from 35.5 mol% for BR11-1^T to 37.4 mol% for OY2-1^T. These values were found to be similar to those for other known species of the genus *Bartonella*: 37 mol% for *Bartonella alsatica* (Heller *et al.*, 1999), 38 mol% for *Bartonella birtlesii* (Bermond *et al.*, 2000), *Bartonella bovis*, *Bartonella capreoli* (Bermond *et al.*, 2002) and *Bartonella tribocorum* (Heller *et al.*, 1998), 40 mol% for *Bartonella japonica*, *Bartonella silvatica* (Inoue *et al.*, 2010) and *Bartonella grahamii* (Birtles *et al.*, 1995) and 41 mol% for *Bartonella doshiae* (Birtles *et al.*, 1995), *Bartonella elizabethae* and *Bartonella henselae* (Daly *et al.*, 1993).

The DNA fragments of the five loci examined were sequenced and the sequence data were compared with those from the type strains of other known species of the genus *Bartonella*. Sequence similarities of the five loci in the strains to the corresponding type strains of species of the genus *Bartonella* are shown in Table 1. The highest similarity values were 94.6 % for *gltA* and 93.4 % for *rpoB*. These values were lower than the cut-off value for discriminating species of the genus *Bartonella* determined by La Scola *et al.* (2003), indicating that the strains studied represent distinct and novel species of the genus *Bartonella*.

The strains OY2-1^T and OY5-1, BR1-1 and BR11-1^T, FN15-2^T and FN18-1 and KS2-1^T and KS7-1 formed distinct clusters from other known species of the genus *Bartonella* by phylogenetic analysis of concatenated sequences of the five loci. In the relationship between species of the genus *Bartonella* and the major mammalian hosts, each of four novel species of the genus *Bartonella* had a unique host, suggesting that these strains have each evolved towards adaptation to a specific environmental and ecological niche and have specialized to optimize with a given host (Fig. 3).

The data support the hypothesis that all of the strains should be classified as distinct species of the genus *Bartonella*.

In addition, the phylogenetic analysis based on deduced amino acid sequences of proteins encoded by the *ftsZ*, *gltA* and *rpoB* genes (Fig. S1) showed that these four novel species of the genus *Bartonella* also formed distinct clades from other known species of the genus *Bartonella* as did the phylogenetic tree derived from the concatenated sequences of the five loci (Fig. 3).

The results in this study strongly indicate that four novel species of the genus *Bartonella* are clearly distinct from other known species of the genus *Bartonella*. We propose the names *Bartonella jaculi* sp. nov., *Bartonella callosciuri* sp. nov., *Bartonella pachyuromydis* sp. nov. and *Bartonella acomydis* sp. nov. for the strains OY2-1^T and OY5-1, BR1-1 and BR11-1^T, FN15-2^T and FN18-1 and KS2-1^T and KS7-1, respectively.

Description of *Bartonella jaculi* sp. nov.

Bartonella jaculi (ja'cu.li. N.L. gen. n. *jaculi* of *Jaculus*, isolated from *Jaculus orientalis*).

After 14 days of incubation on HIA at 35 °C in a moist atmosphere under 5 % CO₂, colonies appear small (1–2 mm in diameter), round, grey–whitish, smooth and umbonate. Electron microscopic examination reveals small bacilli with pili-like structures but without flagella. The cell size is 1.45 µm in length and 0.59 µm in width. The strain does not have oxidase, catalase and urease activities. The strain does not show the ability to reduce nitrate, the ability to produce indole and ability to degrade indoxyl phosphate. The strain does not hydrolyse *p*-nitrophenyl substrates and trehalose. The strain exhibits arylamidase activity towards leucine, methionine, lysine (alkaline as well as acidic), glycine, glycyglycine, arginine, tryptophan and proline. Can be distinguished from other known

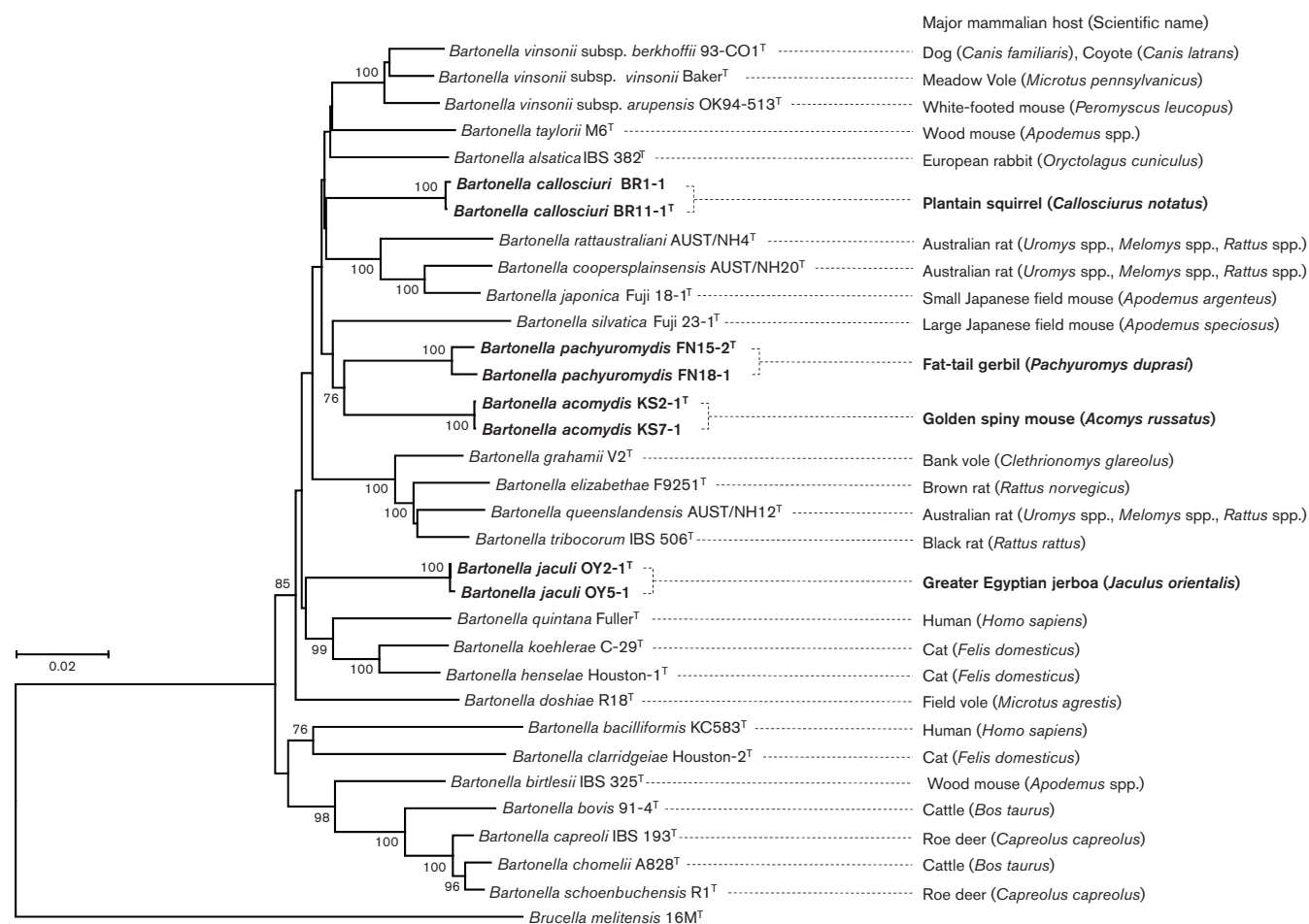


Fig. 3. Phylogenetic relationships based on the concatenated sequences of the 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and the ITS region of the four novel species and other known species of the genus *Bartonella*. The phylogenetic tree was constructed by using the neighbour-joining method with the Jukes–Cantor parameters distance model. The tree was rooted by using *Brucella melitensis* 16M^T as an outgroup. Bootstrap values (percentages of 1000 replications) with greater than 70 % confidence are indicated at the tree nodes. Bar, 0.02 substitutions per nucleotide position.

species of the genus *Bartonella* by the concatenated sequence of its 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and ITS region and has a unique mammalian host, *Jaculus orientalis*.

The type strain is OY2-1^T (=JCM 17712^T=KCTC 23655^T), isolated from the blood of *Jaculus orientalis* mice. The DNA G + C content of the type strain is 37.4 mol% based on the *ftsY* gene sequence. Strain OY5-1 (=JCM 17713=KCTC 23656), isolated from the same source, is a second strain of the species.

Description of *Bartonella callosciuri* sp. nov.

Bartonella callosciuri (cal.lo.sci'u.ri. N.L. gen. n. *callosciuri* of *Callosciurus*, isolated from *Callosciurus notatus*).

After 14 days of incubation on HIA at 35 °C in a moist atmosphere under 5 % CO₂, colonies appear small (1–2 mm in diameter), round, grey–whitish, smooth and umbonate. Electron microscopic examination reveals small bacilli with pili-like structures, but without flagella. The cell size is 1.15 µm in length and 0.56 µm in width. The strain does not have oxidase, catalase and urease activities. The strain does not show the ability to reduce nitrate, the ability to produce indole and the ability to degrade indoxyl phosphate. The strain does not hydrolyse *p*-nitrophenyl substrates and trehalose. The strain exhibits arylamidase activity towards leucine, methionine, lysine (alkaline as well as acidic), glycine, glycyglycine, arginine and tryptophan, but not proline. Can be distinguished from other known species of the genus *Bartonella* by the concatenated sequences of its 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and ITS region and has a unique mammalian host, *Callosciurus notatus*.

The type strain is BR11-1^T (=JCM 17709^T=KCTC 23909^T), isolated from the blood of *Callosciurus notatus* squirrels. The DNA G + C content of the type strain is 35.5 mol% based on the *ftsY* gene sequence. Strain BR1-1 (=JCM 17708=KCTC 23910), isolated from the same source, is a second strain of the species.

Description of *Bartonella pachyuromydis* sp. nov.

Bartonella pachyuromydis (pa.chy.u.ro'my.dis. N.L. gen. n. *pachyuromydis* of *Pachyuromys*, isolated from *Pachyuromys duprasi*).

After 14 days of incubation on HIA at 35 °C in a moist atmosphere under 5 % CO₂, colonies appear small (1–2 mm in diameter), irregular shaped, grey–whitish, smooth and umbonate. Electron microscopic examination reveals small bacilli with pili-like structures but without flagella. The cell size is 1.35 µm in length and 0.59 µm in width. The strain does not have oxidase, catalase and urease activities. The strain does not show the ability to reduce nitrate, the ability to produce indole and the ability to degrade indoxyl phosphate. The strain hydrolyses bis-*p*-nitrophenyl phosphate, but not other *p*-nitrophenyl substrates and trehalose. The strain exhibits arylamidase activity towards leucine, methionine, lysine (alkaline as

well as acidic), glycine, glycyglycine, arginine, tryptophan and proline. Can be distinguished from other known species of the genus *Bartonella* by the concatenated sequences of its 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and ITS region and has a unique mammalian host, *Pachyuromys duprasi*.

The type strain is FN15-2^T (=JCM 17714^T=KCTC 23657^T), isolated from the blood of *Pachyuromys duprasi* mice. The DNA G + C content of the type strain is 35.7 mol% based on the *ftsY* gene sequence. Strain FN18-1 (=JCM 17715=KCTC 23911), isolated from the same source, is a second strain of the species.

Description of *Bartonella acomydis* sp. nov.

Bartonella acomydis (a.co'my.dis. N.L. gen. n. *acomydis* of *Acomys*, isolated from *Acomys russatus*).

After 14 days of incubation on HIA at 35 °C under 5 % CO₂, colonies appear small (1–2 mm in diameter), irregular shaped, grey–whitish, smooth and umbonate. Electron microscopic examination reveals small bacilli with pili-like structures but without flagella. The cell size is 1.39 µm in length and 0.52 µm in width. The strain does not have oxidase, catalase and urease activities. The strain does not show the ability to reduce nitrate, the ability to produce indole and the ability to degrade indoxyl phosphate. The strain hydrolyses bis-*p*-nitrophenyl phosphate but not other *p*-nitrophenyl substrates and trehalose. The strain exhibits arylamidase activity towards leucine, methionine, lysine (alkaline as well as acidic), glycine, glycyglycine, arginine, tryptophan and proline. Can be distinguished from other known species of the genus *Bartonella* by the concatenated sequences of its 16S rRNA, *ftsZ*, *gltA* and *rpoB* genes and ITS region and has a unique mammalian host, *Acomys russatus*.

The type strain is KS2-1^T (=JCM 17706^T=KCTC 23907^T), isolated from the blood of *Acomys russatus* mice. The DNA G + C content of the type strain is 37.2 mol% based on the *ftsY* gene sequence. Strain KS7-1 (=JCM 17707=KCTC 23908), isolated from the same source, is a second strain of the species.

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